

Project 2: Variational Bayes and Gaussian random fields

Deadline: April 7th. Please send your solutions to andrea.riebler@ntnu.no.

General comments:

- This project is compulsory. To be able to take the exam, a reasonable attempt must be made to solve all problems.
- The project report should be formulated as a scientific report. In particular, it should be possible to understand what you have done without reading the questions in this problem text. Moreover, the text in the project report should consist of full sentences and proper punctuation should be used throughout, also in equations! All results you present should be discussed. What can you (and the world) learn from your results? The project text should be written so that it is easy to follow by your fellow students in MA8702. The project report should consist of one (and only one) pdf-file. It should include derivation of formulas that you are using in your implementations. The project report should also include the code you have used to solve the project and the plots you have generated. Associated to the various plots there should be captions explaining the contents of the plots, and in addition all the plots should be referenced, explained and discussed in the main text of the report

The report can be written in English or Norwegian. The project can be done alone or in groups of two or three persons. If you do the project in a group, only one of the individuals in the group should hand in the solution via email to andrea.riebler@ntnu.no. In your solution, specify your (full) names, NOT student or candidate numbers.

1 Variational Bayes for a simple Gaussian model

Consider the hierarchical model

$$\begin{aligned}X_i \mid \mu, \gamma &\sim \mathcal{N}(\mu, 1/\gamma) & i = 1, \dots, N \\ \mu &\sim \mathcal{N}(0, 1/\tau) \\ \gamma &\sim \text{Gamma}(\alpha, \beta),\end{aligned}$$

where γ and τ denote precision (inverse variance) parameters of the Gaussian distribution.

1.1 Joint distribution

Write down the joint distribution

$$p(x, \mu, \gamma)$$

for the observed data $x = (x_1, \dots, x_N)$ and parameters μ and γ .

1.2 Exact posterior distribution

Using Bayes' rule, derive the posterior distribution

$$p(\mu, \gamma \mid x).$$

Show that it belongs to the Normal–Gamma family, and derive expressions for the posterior parameters.

1.3 Variational approximation

Assume a mean-field variational approximation

$$q(\mu, \gamma) = q(\mu)q(\gamma)$$

Derive $q(\mu)$ and $q(\gamma)$ using the VB update rule:

$$\log q(\mu) = \mathbb{E}_\gamma[\log p(x, \mu, \gamma)] + C$$

and

$$\log q(\gamma) = \mathbb{E}_\mu[\log p(x, \mu, \gamma)] + C$$

and show that this will lead to

$$q(\mu) = \mathcal{N}(\nu_q, 1/\tau_q), \quad q(\gamma) = \text{Gamma}(\alpha_q, \beta_q).$$

Give expressions for ν_q , τ_q , α_q and β_q .

1.3.1 Evidence Lower Bound (ELBO)

Write down the ELBO

$$\mathbb{E}_q[\log p(x, \mu, \gamma)] - \mathbb{E}_q[\log q(\mu, \gamma)].$$

Express it in terms of expectations under $q(\mu)$ and $q(\gamma)$, and identify the entropy terms.

1.3.2 Numerical experiment

Generate a dataset $x_1, \dots, x_N$ from a Gaussian distribution using $N = 100$ and where the correct mean is equal to 5 and the correct precision is equal to 1.

The VB algorithm proceeds as follows

1. Update α_q and β_q .
2. Compute $\mathbb{E}(\gamma)$
3. Update τ_q and ν_q .

Until the ELBO converges.

Implement the VB updates using prior parameters $\alpha = 0.01$, $\beta = 0.01$, $\tau = 1E - 6$.

Plot the approximate posterior $q(\mu, \gamma)$ and compare it with the exact posterior. Comment on the differences.

1.3.3 Additional investigation

Experiment with the model and the data set; try changing the prior and the data generating process. What do you observe?

2 Gaussian random fields and Kriging

The purpose of this computer exercise is to give an introduction to parameter estimation and kriging for Gaussian random field models for spatial data. Assume the following observation model on the unit square:

$$y(\mathbf{s}_j) = x(\mathbf{s}_j) + \epsilon_j, \quad j = 1, \dots, N,$$

where $\epsilon_j \sim N(0, \tau^2)$ are independent measurement noise terms. Further, consider a Matérn covariance function for the Gaussian random field $\mathbf{x}(\mathbf{s})$:

$$\text{Cov}(x(\mathbf{s}_i), x(\mathbf{s}_j)) = \Sigma_{i,j} = \sigma^2(1 + \phi h) \exp(-\phi h),$$

where h denotes the Euclidean distance between the two sites $\mathbf{s}_i$ and $\mathbf{s}_j$, and assume the mean increases with east and north coordinates as follows: $\mu_j = \alpha((s_{j1} - 0.5) + (s_{j2} - 0.5))$, for site $\mathbf{s}_j = (s_{j1}, s_{j2})$ on the unit square.

2.1 Simulation

Simulate $N = 200$ random sites in the unit square and plot them. Form the covariance matrix using $\sigma^2 = 1$, $\phi = 10$ and $\tau^2 = 0.05^2$. Take its Cholesky decomposition and simulate dependent zero-mean Gaussian data variables, then add the mean using $\alpha = 1$. Plot your data.

(Hint: The function `rdist` of the package `fields` in R might be useful to compute the distance matrix).

2.2 Parameter estimation

Use the simulated data to estimate the model parameters $\alpha, \sigma^2, \tau^2, \phi$ using maximum likelihood estimation. For this please implement your own optimisation routine, see lecture slides, and show the needed derivations. Iterate between the update for the mean parameter, and updating the covariance parameters as discussed and show how the estimates stabilize. Monitor the likelihood function and plot it for each step of the algorithm to check convergence.

Compare with the true parameter values.

2.3 Kriging

Use the estimated model parameters to perform kriging prediction. Predict variables $\mathbf{x}(s)$, where predictions sites lie on a regular grid of size 25x25 for the unit square. Visualize the Kriging surface and the prediction standard error. Discuss your results.

Add the locations of observations to the plot of the standard deviations and explain the uncertainty pattern you observe.

2.4

Repeat the experiment for a selection of more exotic point structures, for example where observation locations are concentrated on "one-band", "two bands", "circle" or "cross" structures, to see how this affects parameter estimation and kriging estimates.